



## School of Computer Science and Engineering

Winter Semester 2023-2024

Continuous Assessment Test – II

Programme Name & Branch : B.Tech – (BCB/BCE/BCI/BCT/BDS/BKT) SLO : A1+TA1

Course Name & code : BCSE204L – Design and Analysis of Algorithms

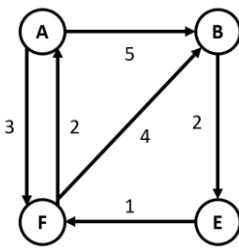
Class Number (s) : ALL

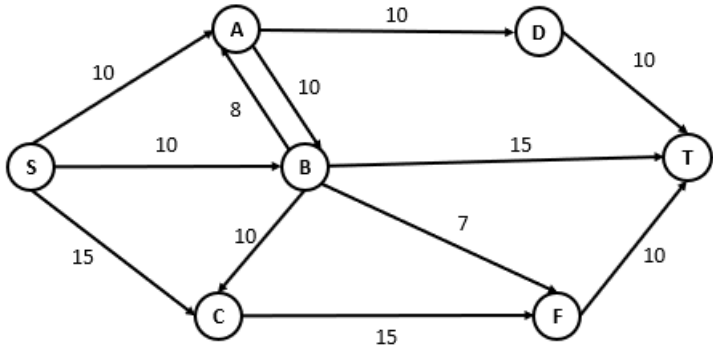
Faculty Name : ALL

Exam Duration : 90 Min.

Maximum Marks: 50

ANSWER ALL THE QUESTIONS(5X10=50 Marks)

| Q.No | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Max Mark |        |        |   |    |   |   |    |   |   |    |   |   |    |   |    |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|--------|---|----|---|---|----|---|---|----|---|---|----|---|----|
| 1    | <p>Solve the Knapsack Problem using FIFOBB, assume knapsack capacity is <math>W = 8</math>. Show how queue is used for node creation in the state space tree.</p> <table><tr><th>Item</th><th>Profit</th><th>Weight</th></tr><tr><td>1</td><td>13</td><td>4</td></tr><tr><td>2</td><td>15</td><td>2</td></tr><tr><td>3</td><td>14</td><td>4</td></tr><tr><td>4</td><td>16</td><td>6</td></tr></table>                                                                                                                                                                                                                  | Item     | Profit | Weight | 1 | 13 | 4 | 2 | 15 | 2 | 3 | 14 | 4 | 4 | 16 | 6 | 10 |
| Item | Profit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Weight   |        |        |   |    |   |   |    |   |   |    |   |   |    |   |    |
| 1    | 13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4        |        |        |   |    |   |   |    |   |   |    |   |   |    |   |    |
| 2    | 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2        |        |        |   |    |   |   |    |   |   |    |   |   |    |   |    |
| 3    | 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4        |        |        |   |    |   |   |    |   |   |    |   |   |    |   |    |
| 4    | 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 6        |        |        |   |    |   |   |    |   |   |    |   |   |    |   |    |
| 2    | <p>Find the existence of a pattern P in the given string S (assign digits A-C as 0-2), using Rabin Karp algorithm. For hash function use Mod 13. Find out how many spurious hits does the algorithm encounter in the <b>Text = ABCBBCABCBAABCCAACB</b> when looking for the pattern <b>Pattern = CCA</b>?</p>                                                                                                                                                                                                                                                                                                          | 10       |        |        |   |    |   |   |    |   |   |    |   |   |    |   |    |
| 3    | <p>Consider a logistics manager tasked with optimizing transportation routes for a delivery company that operates in a city with a complex network of roads. Your goal is to minimize the distance for packages to reach their destinations by finding the shortest paths between all pairs of locations. The transportation route is represented as weighted directed graph given below. Find the shortest paths between all pairs of locations, considering the varying distance, which helps company to delivery operation.</p>  | 10       |        |        |   |    |   |   |    |   |   |    |   |   |    |   |    |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 4 | <p>In water distribution systems, we need to find the maximum amount of water that can be supplied from source S to destination T through a network pipes with capacity limitations. Given a directed graph <math>G=(V,E)</math> representing a water distribution system, where V is set of vertices and E is the set of edges, each edge <math>(u,v)</math> has a capacity <math>c(u,v)</math> representing the maximum water flow that can be supplied through the network pipe. Use Push Relabel algorithm to find the maximum water flow that can be supplied from node S to node T using given graph.</p>  | 10 |
| 5 | <p>Find whether the following line segments intersect or not using cross product.</p> <ol style="list-style-type: none"> <li>L1 :{(1,23) &amp; (10,15)} and L2 :{(4,10) &amp; (6,20)}</li> <li>L3 :{(4,5) &amp; (7,10)} and L4 :{(1,1) &amp; (5,5)}</li> <li>L5 :{(1,1) &amp; (10,10)} and L6 :{(3,3) &amp; (5,5)}</li> <li>L7 :{(1,1) &amp; (10,10)} and L8 :{(5,8) &amp; (3,3)}</li> </ol>                                                                                                                                                                                                                                                                                                       | 10 |



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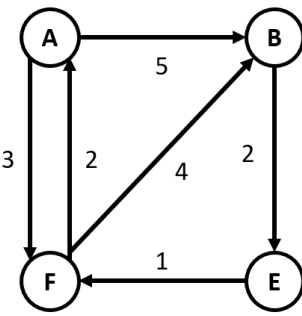
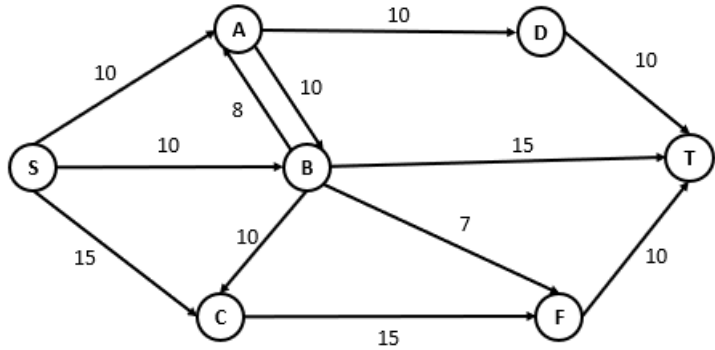
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ANSWER ALL THE QUESTIONS (5X10=50 Marks)

| Q.No | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Module | Max Mark | CO     | BL  |    |   |   |    |   |   |    |   |   |    |   |   |    |     |     |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------|--------|-----|----|---|---|----|---|---|----|---|---|----|---|---|----|-----|-----|
| 1    | <p>Solve the Knapsack Problem using FIFOBB, assume knapsack capacity is <math>W = 8</math>. Show how queue is used for node creation in the state space tree.</p> <table><tr><th>Item</th><th>Profit</th><th>Weight</th></tr><tr><td>1</td><td>13</td><td>4</td></tr><tr><td>2</td><td>15</td><td>2</td></tr><tr><td>3</td><td>14</td><td>4</td></tr><tr><td>4</td><td>16</td><td>6</td></tr></table>                                                                                                                              | Item   | Profit   | Weight | 1   | 13 | 4 | 2 | 15 | 2 | 3 | 14 | 4 | 4 | 16 | 6 | 2 | 10 | CO2 | BL3 |
| Item | Profit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Weight |          |        |     |    |   |   |    |   |   |    |   |   |    |   |   |    |     |     |
| 1    | 13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4      |          |        |     |    |   |   |    |   |   |    |   |   |    |   |   |    |     |     |
| 2    | 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2      |          |        |     |    |   |   |    |   |   |    |   |   |    |   |   |    |     |     |
| 3    | 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4      |          |        |     |    |   |   |    |   |   |    |   |   |    |   |   |    |     |     |
| 4    | 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 6      |          |        |     |    |   |   |    |   |   |    |   |   |    |   |   |    |     |     |
| 2    | <p>Find the existence of a pattern P in the given string S (assign digits A-C as 0-2), using Rabin Karp algorithm. For hash function use Mod 13. Find out how many spurious hits does the algorithm encounter in the <b>Text = ABCBBCABCBAABCCAACB</b> when looking for the pattern <b>Pattern = CCA</b>?</p>                                                                                                                                                                                                                      | 3      | 10       | CO3    | BL2 |    |   |   |    |   |   |    |   |   |    |   |   |    |     |     |
| 3    | <p>Consider a logistics manager tasked with optimizing transportation routes for a delivery company that operates in a city with a complex network of roads. Your goal is to minimize the distance for packages to reach their destinations by finding the shortest paths between all pairs of locations. The transportation route is represented as weighted directed graph given below. Find the shortest paths between all pairs of locations, considering the varying distance, which helps company to delivery operation.</p> | 4      | 10       | CO3    | BL3 |    |   |   |    |   |   |    |   |   |    |   |   |    |     |     |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |    |     |     |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|-----|
|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |    |     |     |
| 4 | <p>In water distribution systems, we need to find the maximum amount of water that can be supplied from source S to destination T through a network pipes with capacity limitations. Given a directed graph <math>G=(V,E)</math> representing a water distribution system, where V is set of vertices and E is the set of edges, each edge <math>(u,v)</math> has a capacity <math>c(u,v)</math> representing the maximum water flow that can be supplied through the network pipe. Use Push Relabel algorithm to find the maximum water flow that can be supplied from node S to node T using given graph.</p>  | 4 | 10 | C03 | BL3 |
| 5 | <p>Find whether the following line segments intersect or not using cross product.</p> <p>e. <math>L1 : \{(1,23) \&amp; (10,15)\}</math> and <math>L2 : \{(4,10) \&amp; (6,20)\}</math></p> <p>f. <math>L3 : \{(4,5) \&amp; (7,10)\}</math> and <math>L4 : \{(1,1) \&amp; (5,5)\}</math></p> <p>g. <math>L5 : \{(1,1) \&amp; (10,10)\}</math> and <math>L6 : \{(3,3) \&amp; (5,5)\}</math></p> <p>a. <math>L7 : \{(1,1) \&amp; (10,10)\}</math> and <math>L8 : \{(5,8) \&amp; (3,3)\}</math></p>                                                                                                                                                                                                     | 5 | 10 | C03 | BL2 |



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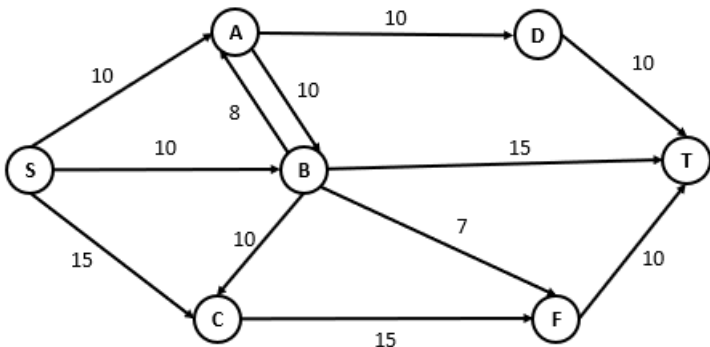
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| Q.No | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Module | Max Mark | CO     | BL |    |   |   |    |   |   |    |   |   |    |   |   |    |     |     |
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| Item | Profit                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Weight |          |        |    |    |   |   |    |   |   |    |   |   |    |   |   |    |     |     |
| 1    | 13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4      |          |        |    |    |   |   |    |   |   |    |   |   |    |   |   |    |     |     |
| 2    | 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2      |          |        |    |    |   |   |    |   |   |    |   |   |    |   |   |    |     |     |
| 3    | 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4      |          |        |    |    |   |   |    |   |   |    |   |   |    |   |   |    |     |     |
| 4    | 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 6      |          |        |    |    |   |   |    |   |   |    |   |   |    |   |   |    |     |     |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |               |    |               |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |             |    |               |   |    |     |     |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------|----|---------------|-----|--------------|---|--|-----|--------------|---|--|-----|--------------|---|--|-----|--------------|---|--|-----|--------------|---|--|-----|-------------|----|---------------|---|----|-----|-----|
|     | <p><math>U = -(13 + 15) = -28</math>, <b>Upper = -28</b><br/><math>\hat{C} = -(28+(8-(4+2))14/4) = -35</math> , Node 2, 3, 4, 5 created<br/><b>Node 2:</b><br/><math>U = -(13 + 15) = -28</math>,<br/><math>\hat{C} = -(28+(8-(4+2))14/4) = -35</math> , Node 6,7,8 created<br/><b>Node 3:</b><br/><math>U = -(15 + 14) = -29</math>, <b>Upper = -29</b><br/><math>\hat{C} = -(29+(8-(2+4))16/2) = -41.33</math> , Node 9,10 created<br/><b>Node 4:</b><br/><math>U = -(14) = -14</math>, <b>Node killed U&gt; Upper</b><br/><b>Node 5:</b><br/><math>U = -(16) = -16</math>, <b>Node killed U&gt; Upper</b><br/><b>Node 6:</b><br/><math>U = -(13 + 15) = -28</math>, <b>Node killed U&gt;Upper</b><br/><math>\hat{C} = -(28+(8-(4+2))14/4) = -35</math> , Node 2, 3, 4, 5 created<br/><b>Node 7:</b><br/><math>U = -(13 + 14) = -27</math>, <b>Node killed U&gt; Upper</b><br/><b>Node 8:</b><br/><math>U = -(13) = -13</math>, <b>Node killed U&gt; Upper</b><br/><math>\hat{C} = -(29+(8-(2+4))16/2) = -41.33</math> , Node 9,10 created<br/><b>Node 9:</b><br/><math>U = -(15 + 14) = -29</math>,<br/><math>\hat{C} = -(29+(8-(2+4))16/2) = -41.33</math> , Node 11 created<br/><b>Node 10:</b><br/><math>U = -(15 + 16) = -31</math>, <b>Upper=-31</b><br/><math>\hat{C} = -(31+(8-(2+6))) = -31</math><br/><b>Node 11:</b><br/><math>U = -(15 + 14) = -29</math>, <b>Node killed U&gt; Upper</b><br/>Answer: Job2 amd Job4 are selected with <b>Maximum profit = 31</b></p> |     |               |    |               |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |             |    |               |   |    |     |     |
| 2   | <p>Find the existence of a pattern P in the given string S (assign digits A-C as 0-2), using Rabin Karp algorithm. For hash function use Mod 13. Find out how many spurious hits does the algorithm encounter in the <b>Text =ABCBBCABCBAABCCAACB</b> when looking for the pattern <b>Pattern = CCA</b>?</p> <p>Ans<br/><math>H(X) = \{P[2]*10^2+P[1]*10^1+P[0]\} \bmod 13</math>.<br/>Assign value A=0, B=1, C=2<br/><math>H(P) = (200+20+0) \bmod 13 = 220 = 12</math>.</p> <p>Text: <b>ABCBBCABCBAABCCAACB</b></p> <table><tr><td>ABC</td><td>12 MOD 13 =</td><td>12</td><td>spurious hits</td></tr><tr><td>BCB</td><td>121 MOD 13 =</td><td>4</td><td></td></tr><tr><td>CBB</td><td>211 MOD 13 =</td><td>3</td><td></td></tr><tr><td>BBC</td><td>112 MOD 13 =</td><td>8</td><td></td></tr><tr><td>BCA</td><td>120 MOD 13 =</td><td>3</td><td></td></tr><tr><td>CAB</td><td>201 MOD 13 =</td><td>6</td><td></td></tr><tr><td>ABC</td><td>12 MOD 13 =</td><td>12</td><td>spurious hits</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ABC | 12 MOD 13 =   | 12 | spurious hits | BCB | 121 MOD 13 = | 4 |  | CBB | 211 MOD 13 = | 3 |  | BBC | 112 MOD 13 = | 8 |  | BCA | 120 MOD 13 = | 3 |  | CAB | 201 MOD 13 = | 6 |  | ABC | 12 MOD 13 = | 12 | spurious hits | 2 | 10 | CO2 | BL6 |
| ABC | 12 MOD 13 =                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 12  | spurious hits |    |               |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |             |    |               |   |    |     |     |
| BCB | 121 MOD 13 =                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4   |               |    |               |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |             |    |               |   |    |     |     |
| CBB | 211 MOD 13 =                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3   |               |    |               |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |             |    |               |   |    |     |     |
| BBC | 112 MOD 13 =                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 8   |               |    |               |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |             |    |               |   |    |     |     |
| BCA | 120 MOD 13 =                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3   |               |    |               |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |             |    |               |   |    |     |     |
| CAB | 201 MOD 13 =                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 6   |               |    |               |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |             |    |               |   |    |     |     |
| ABC | 12 MOD 13 =                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 12  | spurious hits |    |               |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |              |   |  |     |             |    |               |   |    |     |     |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |    |     |     |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|-----|
|   | <div> <div> <div>BCB</div> <div>121 MOD 13 =</div> <div>4</div> </div> <div> <div>CBA</div> <div>210 MOD 13 =</div> <div>2</div> </div> <div> <div>BAA</div> <div>100 MOD 13 =</div> <div>9</div> </div> <div> <div>AAA</div> <div>0 MOD 13 =</div> <div>0</div> </div> <div> <div>AAB</div> <div>1 MOD 13 =</div> <div>1</div> </div> <div> <div>ABC</div> <div>12 MOD 13 =</div> <div>12</div> </div> <div> <div>BCC</div> <div>122 MOD 13 =</div> <div>5</div> </div> <div> <div>CCA</div> <div>220 MOD 13 =</div> <div>12</div> </div> <div> <div>CAA</div> <div>200 MOD 13 =</div> <div>5</div> </div> <div> <div>AAC</div> <div>2 MOD 13 =</div> <div>2</div> </div> <div> <div>ACB</div> <div>21 MOD 13 =</div> <div>8</div> </div> </div> <div> <div>spurious hits</div> <div>Hit</div> </div>                                                    |   |    |     |     |
|   | Total number of spurious hits is 3 for the patten CCA.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |    |     |     |
| 3 | <p>Consider a logistics manager tasked with optimizing transportation routes for a delivery company that operates in a city with a complex network of roads. Your goal is to minimize the distance for packages to reach their destinations by finding the shortest paths between all pairs of locations. The transportation route is represented as weighted directed graph given below. Find the shortest paths between all pairs of locations, considering the varying distance, which helps company to delivery operation.</p> <div> <div> <div> <div> <div>A</div> <div>B</div> </div> <div> <div> <div>3</div> <div>2</div> </div> <div> <div> <div>5</div> <div>4</div> </div> <div> <div>2</div> <div>1</div> </div> </div> </div> </div> </div> <div> <div> <div> <div> <div>F</div> <div>E</div> </div> </div> </div> </div> </div> <p>Ans:</p> | 3 | 10 | CO3 | BL6 |

|   | <div><div><div><math>A^0</math></div><table><tr><th></th><th>a</th><th>b</th><th>e</th><th>f</th></tr><tr><th>a</th><td>0</td><td>5</td><td><math>\infty</math></td><td>3</td></tr><tr><th>b</th><td><math>\infty</math></td><td>0</td><td>2</td><td><math>\infty</math></td></tr><tr><th>e</th><td><math>\infty</math></td><td><math>\infty</math></td><td>0</td><td>1</td></tr><tr><th>f</th><td>2</td><td>4</td><td><math>\infty</math></td><td>0</td></tr></table></div><div><div><math>A^A</math></div><table><tr><th></th><th>a</th><th>b</th><th>e</th><th>f</th></tr><tr><th>a</th><td>0</td><td>5</td><td><math>\infty</math></td><td>3</td></tr><tr><th>b</th><td><math>\infty</math></td><td>0</td><td>2</td><td><math>\infty</math></td></tr><tr><th>e</th><td><math>\infty</math></td><td><math>\infty</math></td><td>0</td><td>1</td></tr><tr><th>f</th><td>2</td><td>4</td><td><math>\infty</math></td><td>0</td></tr></table></div><div><div><math>A^B</math></div><table><tr><th></th><th>a</th><th>b</th><th>e</th><th>f</th></tr><tr><th>a</th><td>0</td><td>5</td><td>7</td><td>3</td></tr><tr><th>b</th><td><math>\infty</math></td><td>0</td><td>2</td><td><math>\infty</math></td></tr><tr><th>e</th><td><math>\infty</math></td><td><math>\infty</math></td><td>0</td><td>1</td></tr><tr><th>f</th><td>2</td><td>4</td><td>6</td><td>0</td></tr></table></div><div><div><math>A^E</math></div><table><tr><th></th><th>a</th><th>b</th><th>e</th><th>f</th></tr><tr><th>a</th><td>0</td><td>5</td><td>7</td><td>3</td></tr><tr><th>b</th><td><math>\infty</math></td><td>0</td><td>2</td><td>3</td></tr><tr><th>e</th><td><math>\infty</math></td><td><math>\infty</math></td><td>0</td><td>1</td></tr><tr><th>f</th><td>2</td><td>4</td><td>6</td><td>0</td></tr></table></div><div><div><math>A^F</math></div><table><tr><th></th><th>a</th><th>b</th><th>e</th><th>f</th></tr><tr><th>a</th><td>0</td><td>5</td><td>7</td><td>3</td></tr><tr><th>b</th><td>5</td><td>0</td><td>2</td><td>3</td></tr><tr><th>e</th><td>3</td><td>5</td><td>0</td><td>1</td></tr><tr><th>f</th><td>2</td><td>4</td><td>6</td><td>0</td></tr></table></div></div> |          | a        | b        | e   | f | a | 0 | 5 | $\infty$ | 3 | b | $\infty$ | 0 | 2 | $\infty$ | e | $\infty$ | $\infty$ | 0 | 1 | f | 2 | 4 | $\infty$ | 0 |  | a | b | e | f | a | 0 | 5 | $\infty$ | 3 | b | $\infty$ | 0 | 2 | $\infty$ | e | $\infty$ | $\infty$ | 0 | 1 | f | 2 | 4 | $\infty$ | 0 |  | a | b | e | f | a | 0 | 5 | 7 | 3 | b | $\infty$ | 0 | 2 | $\infty$ | e | $\infty$ | $\infty$ | 0 | 1 | f | 2 | 4 | 6 | 0 |  | a | b | e | f | a | 0 | 5 | 7 | 3 | b | $\infty$ | 0 | 2 | 3 | e | $\infty$ | $\infty$ | 0 | 1 | f | 2 | 4 | 6 | 0 |  | a | b | e | f | a | 0 | 5 | 7 | 3 | b | 5 | 0 | 2 | 3 | e | 3 | 5 | 0 | 1 | f | 2 | 4 | 6 | 0 |  |  |  |  |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----|---|---|---|---|----------|---|---|----------|---|---|----------|---|----------|----------|---|---|---|---|---|----------|---|--|---|---|---|---|---|---|---|----------|---|---|----------|---|---|----------|---|----------|----------|---|---|---|---|---|----------|---|--|---|---|---|---|---|---|---|---|---|---|----------|---|---|----------|---|----------|----------|---|---|---|---|---|---|---|--|---|---|---|---|---|---|---|---|---|---|----------|---|---|---|---|----------|----------|---|---|---|---|---|---|---|--|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|--|--|--|
|   | a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | b        | e        | f        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| a | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5        | $\infty$ | 3        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| b | $\infty$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0        | 2        | $\infty$ |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| e | $\infty$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\infty$ | 0        | 1        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| f | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4        | $\infty$ | 0        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
|   | a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | b        | e        | f        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| a | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5        | $\infty$ | 3        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| b | $\infty$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0        | 2        | $\infty$ |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| e | $\infty$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\infty$ | 0        | 1        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| f | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4        | $\infty$ | 0        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
|   | a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | b        | e        | f        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| a | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5        | 7        | 3        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| b | $\infty$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0        | 2        | $\infty$ |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| e | $\infty$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\infty$ | 0        | 1        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| f | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4        | 6        | 0        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
|   | a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | b        | e        | f        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| a | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5        | 7        | 3        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| b | $\infty$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0        | 2        | 3        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| e | $\infty$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\infty$ | 0        | 1        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| f | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4        | 6        | 0        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
|   | a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | b        | e        | f        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| a | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5        | 7        | 3        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| b | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0        | 2        | 3        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| e | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5        | 0        | 1        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| f | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4        | 6        | 0        |     |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |
| 4 | <p>In water distribution systems, we need to find the maximum amount of water that can be supplied from source S to destination T through a network pipes with capacity limitations. Given a directed graph <math>G=(V,E)</math> representing a water distribution system, where V is set of vertices and E is the set of edges, each edge <math>(u,v)</math> has a capacity <math>c(u,v)</math> representing the maximum water flow that can be supplied through the network pipe. Use Push Relabel algorithm to find the maximum water flow that can be supplied from node S to node T using given graph.</p> <div></div> <p>Ans:</p> <ul style="list-style-type: none"><li>Max flow 30</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4        | 10       | C03      | BL6 |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |          |   |   |          |   |   |          |   |          |          |   |   |   |   |   |          |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |          |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |          |   |   |   |   |          |          |   |   |   |   |   |   |   |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |  |



|   |                                                                                                                                                                                                                                                                                                                                                                                              |   |    |     |     |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|-----|
|   | <p>The first diagram shows the initial state with H values: S=7, A=1, B=0, C=0, D=1, F=0, T=0. E values are all 0. The second diagram shows an augmenting path S→B→T with flow 10, updating H and E values. The third diagram shows two augmenting paths: S→B→T (flow 10) and S→C→F→T (flow 5), further updating the flow network.</p>                                                       |   |    |     |     |
| 5 | <p>Find whether the following line segments intersect or not using cross product.</p> <ol style="list-style-type: none"> <li>L1 :{(1,23) &amp; (10,15)} and L2 :{(4,10) &amp; (6,20)}</li> <li>L3 :{(4,5) &amp; (7,10)} and L4 :{(1,1) &amp; (5,5)}</li> <li>L5 :{(1,1) &amp; (10,10)} and L6 :{(3,3) &amp; (5,5)}</li> <li>L7 :{(1,1) &amp; (10,10)} and L8 :{(5,8) &amp; (3,3)}</li> </ol> | 4 | 10 | C03 | BL6 |

Ans:

$$\begin{aligned} \text{a. } P11 \times (\overline{P12 P21}) \\ = \begin{vmatrix} (4-1) & (10-1) \\ (10-23) & (15-23) \end{vmatrix} = \begin{vmatrix} 3 & 9 \\ -13 & -8 \end{vmatrix} = +ve \end{aligned}$$

$$\begin{aligned} P11 \times (\overline{P12 P22}) \\ = \begin{vmatrix} (6-1) & (10-1) \\ (20-23) & (15-23) \end{vmatrix} = \begin{vmatrix} 5 & 9 \\ -3 & -8 \end{vmatrix} = -ve \end{aligned}$$

*L1 and L2 intersect*

$$\begin{aligned} \text{b. } P31 \times (\overline{P32 P41}) \\ = \begin{vmatrix} (1-4) & (7-4) \\ (1-5) & (10-5) \end{vmatrix} = \begin{vmatrix} -3 & 3 \\ -4 & 5 \end{vmatrix} = -ve \end{aligned}$$

$$\begin{aligned} P31 \times (\overline{P32 P42}) \\ = \begin{vmatrix} (5-4) & (7-4) \\ (5-5) & (10-5) \end{vmatrix} = \begin{vmatrix} 1 & 3 \\ 0 & 5 \end{vmatrix} = +ve \end{aligned}$$

*L3 and L4 intersect*

$$\begin{aligned} \text{c. } P51 \times (\overline{P52 P61}) \\ = \begin{vmatrix} (3-1) & (10-1) \\ (3-1) & (10-1) \end{vmatrix} = \begin{vmatrix} 2 & 9 \\ 2 & 9 \end{vmatrix} = 0 \end{aligned}$$

$$\begin{aligned} P51 \times (\overline{P52 P62}) \\ = \begin{vmatrix} (5-1) & (10-1) \\ (5-1) & (10-1) \end{vmatrix} = \begin{vmatrix} 4 & 9 \\ 4 & 9 \end{vmatrix} = 0 \end{aligned}$$

Bounding Box for L5

$$\widehat{P51} = \{\min(1, 10), \min(1, 10)\} = \{(1, 1)\}$$

$$\widehat{P52} = \{\max(1, 10), \max(1, 10)\} = \{(10, 10)\}$$

Bounding Box for L6

$$\widehat{P61} = \{\min(3, 5), \min(3, 5)\} = \{(3, 3)\}$$

$$\widehat{P62} = \{\max(3, 5), \max(3, 5)\} = \{(5, 5)\}$$

If  $\{P51, P52, \widehat{P51}, \widehat{P52}\}$  and  $\{P61, P62, \widehat{P61}, \widehat{P62}\}$

$(10 > 3)$  and  $(5 > 1)$  and  $(10 > 3)$  and  $(5 > 1)$  hence

*L5 and L6 intersect*

$$\begin{aligned} \text{d. } P71 \times (\overline{P72 P81}) \\ = \begin{vmatrix} (5-1) & (10-1) \\ (8-1) & (10-1) \end{vmatrix} = \begin{vmatrix} 4 & 9 \\ 7 & 9 \end{vmatrix} = -ve \end{aligned}$$

$$\begin{aligned} P71 \times (\overline{P72 P82}) \\ = \begin{vmatrix} (3-1) & (10-1) \\ (3-1) & (10-1) \end{vmatrix} = \begin{vmatrix} 2 & 9 \\ 2 & 9 \end{vmatrix} = 0 \end{aligned}$$

*L1 and L2 intersect*